Curriculum Vitae

Ali Baheri | akbeme@rit.edu | alibaheri.github.io

Assistant Professor, Department of Mechanical Engineering, Rochester Institute of Technology

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Last updated: September 3, 2026

Research Areas

Safe learning; reinforcement learning; robotics and autonomous systems; responsible and safe AI.

Academic Appointments

2023–present	**Assistant Professor (tenure-track)**, Rochester Institute of Technology Department of Mechanical Engineering; Director, Safe AI Lab
2022–2023	**Visiting Scholar**, Stanford University
2019–2022	**Assistant Professor (research-track)**, West Virginia University
2018–2019	**Postdoctoral Fellow**, University of Michigan, Ann Arbor

Education

2015–2018	**Ph.D., Mechanical Engineering**, University of North Carolina at Charlotte Dissertation research: machine learning, dynamics, and control
2012–2014	**M.S.**, University of Louisiana at Lafayette Mechanical Engineering: systems, dynamics, and control
2002–2006	**B.S.**, Sharif University of Technology Mechanical Engineering

Honors and Awards

2025	Nominee, **Outstanding Teaching Award** (Eisenhart Awards), Rochester Institute of Technology
2023	**AAAI New Faculty Highlights**, AAAI-23 (invited program for selected early-career faculty)
2022	National Science Foundation EPSCoR Fellowship
2018	Ford-Michigan Postdoctoral Fellowship

External Research Funding

	Approximately $1.47M as single or lead PI; senior personnel on a $2.2M NSF National Research Traineeship.
2026–2029	**NSA / NCAE-C: Red-Teaming and Assurance for Geometric AI in Defense Autonomy**, *$719K*, Single PI
2022–2024	**NSF: RII: Safety Validation of Autonomous Systems from Multiple Sources of Information**, ≈*$200K*, Single PI
2021–2026	**NSF: NRT-AI: AWARE-AI: AWAREness for Sensing Humans Responsibly with AI**, *$2.2M*, Senior Personnel
2021–2023	**Federal Aviation Administration: Safety Verification Framework for Learning-based Aviation Systems (SVF-LAS)**, *$400K*, Lead PI

2021–2022	**NASA: Fault Diagnosis for Safety-Critical Autonomous Systems using Reinforcement Learning**, *$100K*, Lead PI
2021–2022	**NASA WV Space Grant Consortium: Black-Box Verification of Autonomous Systems Using Modular Reinforcement Learning**, ≈*$30K*, Single PI
2020–2021	**NASA WV Space Grant Consortium: Robust Autonomy Through Experimentally Infused Decision Making with the Application to Planetary Mars Rover**, ≈*$23K*, Single PI

Internal Funding

2025	**ConceptCrystal: AI-Powered Concept Visualization**, *RIT Provost's Learning Innovations Grant, $2K*, Single PI
2024	**HARMONY: Hierarchical Anchored Representation for Multidomain Operability in Next-gen Systems**, *RIT Research Office, $5K*, Single PI
2024	**Provost's Leadership Opportunity Grants**, *Rochester Institute of Technology, $2K*
2021–2022	**Verification of Multi-Agent Autonomous Planning and Control**, *West Virginia University Research Office Program, ≈$25K*, Single PI

Publications

Peer-Reviewed Journal Articles

[**J22**] **Ali Baheri**, How Much Do Your Models Disagree? Adaptive MPC Safety from Ensemble Uncertainty. *IEEE Open Journal of Control Systems*, 2026. (Special Section: Intersection of Machine Learning with Control)

[**J21**] Z. Shahrooei, **Ali Baheri**, Blending Optimism and Pessimism via Wasserstein Barycenters for Continuous Control. *IEEE Control Systems Letters*, 2026.

[**J20**] M. Amiri Shahbazi, **Ali Baheri**, Heterogeneous-Horizon Conformal Ensembles for Online Prediction Under Distribution Shift: An Empirical Comparison with Strongly-Adaptive Methods. *Forecasting*, accepted, 2026.

[**J19**] C. Nau, K. McConky, C. Alm, R. Bailey, **Ali Baheri**, P. Sankaran, M. Sudit, Fairness Without Discrimination: Individually Fair Outcomes in the Kidney Exchange Problem. *Decision Analysis*, 2026.

[**J18**] Z. Shahrooei, M. Kochenderfer, **Ali Baheri**, Efficient Counterexample Generation for Control Systems Using Multi-Fidelity Bayesian Optimization. *IEEE Access*, Vol. 14, pp. 61902–61922, 2026.

[**J17**] M. Amiri Shahbazi, **Ali Baheri**, N. Azadeh-Fard, Hospital and Regional Effects on Length of Stay: A Multilevel Modeling Approach. *IISE Transactions on Healthcare Systems Engineering*, 2026.

[**J16**] M. Amiri Shahbazi, **Ali Baheri**, N. Azadeh-Fard, A Hierarchical Conformal Framework for Uncertainty-Aware Length of Stay Prediction in Multi-Hospital Settings. *Scientific Reports*, 2026.

[**J15**] **Ali Baheri**, P. Wei, Multi-Fidelity Temporal Reasoning: A Stratified Logic for Cross-Scale System Specifications. *Logics*, Vol. 3, No. 2, 2025.

[**J14**] **Ali Baheri**, M. Amiri Shahbazi, Conformal Prediction Across Scales: Finite-Sample Coverage with Hierarchical Efficiency. *Results in Applied Mathematics*, Vol. 26, 100589, 2025.

[J13] **Ali Baheri**, Distributionally Robust Lyapunov-Barrier Networks for Safe and Stable Control Under Uncertainty. *Results in Control and Optimization*, Vol. 19, 2025.

[J12] **Ali Baheri**, Multilevel Constrained Bandits: A Hierarchical Upper Confidence Bound Approach with Safety Guarantees. *Mathematics*, 13(1), 149, 2025.

[J11] K. Hayes, M. Fouts, **Ali Baheri**, D. Mebane, Forward Variable Selection Enables Fast and Accurate Dynamic System Identification with Karhunen-Loève Decomposed Gaussian Processes. *PLOS ONE*, 2024.

[J10] J. Yancosek, **Ali Baheri**, BEACON: A Bayesian Evolutionary Approach for Counterexample Generation of Control Systems. *IEEE Access*, 2024.

[J9] P. Razzaghi, A. Tabrizian, W. Guo, S. Chen, A. Taye, E. Thompson, A. Bregeon, **Ali Baheri**, P. Wei, A Survey on Reinforcement Learning in Aviation Applications. *Engineering Applications of Artificial Intelligence*, Vol. 135, 2024.

[J8] L. Yifru, **Ali Baheri**, Concurrent Learning of Control Policy and Unknown Safety Specifications in Reinforcement Learning. *IEEE Open Journal of Control Systems*, Vol. 3, pp. 266–281, 2024.

[J7] **Ali Baheri**, Exploring the Role of Simulator Fidelity in the Safety Validation of Learning-Enabled Autonomous Systems. *AI Magazine*, Vol. 44, pp. 453–459, 2023.

[J6] **Ali Baheri**, Safe Reinforcement Learning with Mixture Density Network, with Application to Autonomous Driving. *Results in Control and Optimization*, Vol. 6, 2022.

[J5] **Ali Baheri**, C. Vermillion, Combined Plant and Controller Design Using Batch Bayesian Optimization: A Case Study in Airborne Wind Energy Systems. *ASME Journal of Dynamic Systems, Measurement, and Control*, Vol. 141, Issue 9, 2019.

[J4] S. Bin-Karim, A. Bafandeh, **Ali Baheri**, C. Vermillion, Spatiotemporal Optimization Through Gaussian Process-Based Model Predictive Control: A Case Study in Airborne Wind Energy. *IEEE Transactions on Control Systems Technology*, Vol. 27, Issue 2, pp. 798–805, 2019.

[J3] **Ali Baheri**, P. Ramaprabhu, C. Vermillion, Iterative 3D Layout Optimization and Parametric Trade Study for a Reconfigurable Ocean Current Turbine Array Using Bayesian Optimization. *Renewable Energy*, Vol. 127, pp. 1052–1063, 2018.

[J2] A. Bafandeh, S. Bin-Karim, **Ali Baheri**, C. Vermillion, A Comparative Assessment of Hierarchical Control Structures for Spatiotemporally Varying Systems, with Application to Airborne Wind Energy. *Control Engineering Practice*, Vol. 74, pp. 71–83, 2018.

[J1] **Ali Baheri**, S. Bin-Karim, A. Bafandeh, C. Vermillion, Real-Time Control Using Bayesian Optimization: A Case Study in Airborne Wind Energy Systems. *Control Engineering Practice*, Vol. 69, pp. 131–140, 2017.

Refereed Conference Proceedings

[C22] **Ali Baheri**, Wasserstein Stability of Contracting Flows: Effective Rates, Euler Self-Correction, and Noise Tightening. *IEEE Conference on Decision and Control (CDC)*, 2026.

[C21] **Ali Baheri**, When Does Multi-Agent Language-Model Debate Converge? A Bifurcation on the Probability Simplex. *Conference on Complex Systems (CCS)*, accepted, 2026.

[C20] **Ali Baheri**, L. Lindemann, Metriplectic Conditional Flow Matching for Structure-Preserving Dynamics Learning. *European Control Conference (ECC)*, 2026.

[C19] S. Karpoora Sundara Pandian, **Ali Baheri**, Density-Ratio Weighted Behavioral Cloning: Learning Control Policies from Corrupted Datasets. *American Control Conference (ACC)*, 2026.

[C18] **Ali Baheri**, A. Vahid, Geometry-Aware Decentralized Sinkhorn for Wasserstein Barycenters. *American Control Conference (ACC)*, 2026.

- **[C17]** D. Millard, **Ali Baheri**, Can Optimal Transport Improve Federated Inverse Reinforcement Learning? *Learning for Dynamics & Control Conference (L4DC)*, 2026.
- **[C16]** **Ali Baheri**, Logic-Guided Vector Fields for Constrained Generative Modeling. *3rd International Conference on Neuro-Symbolic Systems (NeuS)*, 2026. **(Spotlight Talk)**
- **[C15]** **Ali Baheri**, C. Alm, Hierarchical Neuro-Symbolic Decision Transformer. *19th International Conference on Neurosymbolic Learning and Reasoning (NeSy)*, 2025.
- **[C14]** **Ali Baheri**, Z. Shahrooei, C. Salgarkar, WAVE: Wasserstein Adaptive Value Estimation for Actor-Critic Reinforcement Learning. *Learning for Dynamics & Control Conference (L4DC)*, 2025.
- **[C13]** **Ali Baheri**, Safety Validation of Learning-Based Autonomous Systems: A Multi-Fidelity Approach. *Proceedings of the AAAI Conference on Artificial Intelligence (New Faculty Highlights)*, Vol. 37, No. 13, p. 15432, 2023.
- **[C12]** Z. Shahrooei, M. Kochenderfer, **Ali Baheri**, Falsification of Learning-Based Controllers through Multi-Fidelity Bayesian Optimization. *European Control Conference (ECC)*, Bucharest, Romania, 2023.
- **[C11]** **Ali Baheri**, H. Ren, B. Johnson, P. Razzaghi, P. Wei, A Verification Framework for Certifying Learning-Based Safety-Critical Aviation Systems. *AIAA*, Chicago, IL, 2022.
- **[C10]** **Ali Baheri**, Safe Reinforcement Learning with Mixture Density Network: A Case Study in Autonomous Highway Driving. *Robotics: Science and Systems (RSS)*, Corvallis, OR, 2020.
- **[C9]** **Ali Baheri**, S. Nageshrao, I. Kolmanovsky, A. Girard, E. Tseng, D. Filev, Deep Reinforcement Learning with Enhanced Safety for Autonomous Highway Driving. *31st IEEE Intelligent Vehicles Symposium (IV)*, Las Vegas, NV, 2020.
- **[C8]** **Ali Baheri**, I. Kolmanovsky, A. Girard, E. Tseng, D. Filev, Vision-Based Autonomous Driving: A Model Learning Approach. *American Control Conference (ACC)*, Denver, CO, 2020.
- **[C7]** **Ali Baheri**, C. Vermillion, Waypoint Optimization Using Bayesian Optimization: A Case Study in Airborne Wind Energy Systems. *American Control Conference (ACC)*, Denver, CO, 2020.
- **[C6]** **Ali Baheri**, J. Deese, C. Vermillion, Combined Plant and Controller Design Using Bayesian Optimization: A Case Study in Airborne Wind Energy Systems. *ASME Dynamic Systems and Control Conference (DSCC)*, Tysons Corner, VA, 2017.
- **[C5]** **Ali Baheri**, P. Ramaprabhu, C. Vermillion, Iterative In-Situ 3D Layout Optimization of a Reconfigurable Ocean Current Turbine Array Using Bayesian Optimization. *ASME Dynamic Systems and Control Conference (DSCC)*, Tysons Corner, VA, 2017.
- **[C4]** **Ali Baheri**, C. Vermillion, Altitude Optimization of Airborne Wind Energy Systems: A Bayesian Optimization Approach. *American Control Conference (ACC)*, Seattle, WA, 2017.
- **[C3]** **Ali Baheri**, J. Vaughan, Concurrent Design of Unity-Magnitude Input Shapers and Proportional-Derivative Feedback Controllers. *American Control Conference (ACC)*, Chicago, IL, 2015.
- **[C2]** **Ali Baheri**, J. Vaughan, Robust Concurrent Design of Input and Proportional-Derivative Feedback Controllers. *International Symposium on Flexible Automation (ISFA)*, Awaji-Island, Japan, 2014.
- **[C1]** **Ali Baheri**, J. Vaughan, Concurrent Command and Mechanical System Design to Limit Transient and Residual Vibration. *International Conference on Motion and Vibration Control (MoViC)*, Sapporo, Japan, 2014.

Workshop Papers and Extended Abstracts

[W18] A. Naghdi, **Ali Baheri**, Flow-Corrected Thompson Sampling for Non-Stationary Contextual Bandits. *Continual RL workshop, RLC*, 2026. **(Oral Presentation)**

[W17] M. Amiri Shahbazi, **Ali Baheri**, Geometry-Aware Uncertainty Quantification via Conformal Prediction on Manifolds. *Epistemic Intelligence in Machine Learning (EIML) workshop, ICML*, 2026. **(Spotlight Talk)**

[W16] **Ali Baheri**, Geometry-Grounded Flow Matching on Compact Manifolds. *Geometry-grounded Representation Learning and Generative Modeling (GRaM) workshop, ICLR*, 2026.

[W15] **Ali Baheri**, What Does a Neural PDE Solver Really Learn? A Residual-Spectrum Diagnostic. *AI and Partial Differential Equations (AI&PDE) workshop, ICLR*, 2026.

[W14] C. Salgarkar, **Ali Baheri**, When Distance Matters: Wasserstein Trust Regions for Multi-Agent Coordination. *Multi-Agent Path Finding workshop, AAAI*, 2026. **(Oral Presentation)**

[W13] M. Amiri Shahbazi, **Ali Baheri**, N. Azadeh-Fard, Adaptive Conformal Prediction via Bayesian Uncertainty Weighting for Hierarchical Healthcare Data. *SECURE-AI4H workshop, AAAI*, 2026.

[W12] **Ali Baheri**, Geometry-Aware Backdoor Attacks: Leveraging Curvature in Hyperbolic Embeddings. *Non-Euclidean Foundation Models and Geometric Learning workshop, NeurIPS*, 2025.

[W11] **Ali Baheri**, L. Lindemann, Metriplectic Conditional Flow Matching for Dissipative Dynamics. *Dynamics at the Frontiers of Optimization, Sampling, and Games (DynaFront) workshop, NeurIPS*, 2025.

[W10] **Ali Baheri**, Wasserstein-Barycenter Consensus for Cooperative Multi-Agent Reinforcement Learning. *Multi-Agent Systems in the Era of Foundation Models workshop, ICML*, 2025.

[W9] **Ali Baheri**, Implicit Constraint-Aware Off-Policy Correction for Offline Reinforcement Learning. *Out-of-Distribution Generalization in Robotics workshop, RSS*, 2025.

[W8] Z. Shahrooei, **Ali Baheri**, Optimal Transport-Assisted Risk-Sensitive Q-Learning. *Towards Safe Autonomy: Emerging Requirements, Definitions, and Methods workshop, RSS*, 2024.

[W7] **Ali Baheri**, C. Alm, LLMs-Augmented Contextual Bandit. *Optimal Transport and Machine Learning workshop, NeurIPS*, 2023.

[W6] **Ali Baheri**, Understanding Reward Ambiguity Through Optimal Transport Theory in Inverse Reinforcement Learning. *Optimal Transport and Machine Learning workshop, NeurIPS*, 2023.

[W5] **Ali Baheri**, Risk-Aware Reinforcement Learning Through Optimal Transport Theory. *3rd RL-CONFORM workshop, IROS*, 2023.

[W4] **Ali Baheri**, Policy Refinement with Human Feedback for Safe Reinforcement Learning. *Bridging the Gap Between AI Planning and Reinforcement Learning (PRL) workshop, ICAPS*, 2023.

[W3] L. Yifru, **Ali Baheri**, Joint Learning of Policy with Unknown Temporal Constraints for Safe Reinforcement Learning. *Bridging the Gap Between AI Planning and Reinforcement Learning (PRL) workshop, ICAPS*, 2023.

[W2] **Ali Baheri**, S. Nageshrao, I. Kolmanovsky, A. Girard, E. Tseng, D. Filev, Deep Q-Learning with Dynamically-Learned Safety Module: A Case Study in Autonomous Driving. *NeurIPS*, Vancouver, Canada, 2019.

[**W1**]	**Ali Baheri**, C. Vermillion, Context-Dependent Bayesian Optimization in Real-Time Optimal Control: A Case Study in Airborne Wind Energy Systems. *NIPS Workshop on Bayesian Optimization*, Long Beach, CA, 2017.

Preprints

[**P5**]	**Ali Baheri**, Step-Level Hallucination Detection via Hidden-State Trajectory Geometry. *Manuscript under review*, 2026.
[**P4**]	D. Millard, C. Alm, R. Ali, P. Shi, **Ali Baheri**, Federated Distributional Reinforcement Learning with Distributional Critic Regularization, 2026. *arXiv:2603.17820*
[**P3**]	**Ali Baheri**, M. Kochenderfer, The Synergy Between Optimal Transport Theory and Multi-Agent Reinforcement Learning, 2024. *arXiv:2401.10949*
[**P2**]	**Ali Baheri**, M. Kochenderfer, Joint Falsification and Fidelity Settings Optimization for Validation of Safety-Critical Systems: A Theoretical Analysis, 2023. *arXiv:2305.06111*
[**P1**]	S. Jacobs, R. Butts, **Ali Baheri**, Y. Gu, G. Pereira, A Framework for Controlling Multi-Robot Systems Using Bayesian Optimization and Linear Combination of Vectors, 2022. *arXiv:2203.12416*

Invited Talks and Presentations

Mar 2024	Evolving AI Decision-Making: From Safe Reinforcement Learning to Intelligent Systems with Language Models, RIT Center for Human-aware AI (CHAI) Seminar
Feb 2023	On the Role of Fidelity in the Safety Evaluation of Learning-Based Autonomous Systems, RIT Graduate Seminar
Feb 2023	On the Role of Fidelity in the Safety Evaluation of Learning-Based Autonomous Systems, AAAI-23 New Faculty Highlights Program
May 2022	Safety Verification of Autonomous Systems: A Multi-Fidelity Reinforcement Learning Approach, ICRA 2022 Workshop on the Verification of Autonomous Systems (VAS)
Apr 2022	Safe Decision Making in Evolving Environments for Safety-Critical Autonomous Systems, University of North Texas
Mar 2022	Lessons from Safe Learning and Safety Validation Research for Autonomous Systems in the Wild, Rochester Institute of Technology
Aug 2020	Safe Learning in Autonomous Driving, Ford Motor Company
Feb 2020	Safe and Human-like Decision Making for Autonomous Systems, University of New Mexico
Nov 2018	Guest Invited Lecture, Deep Reinforcement Learning, University of North Carolina at Charlotte

Teaching and Mentoring

Teaching Interests

Dynamics and control; robotics and autonomous systems; reinforcement learning; machine learning for physical systems; cyber-physical systems; formal and statistical methods for autonomy assurance.

Courses Developed

RIT	**Understanding Reinforcement Learning** (graduate), Fall 2024, Fall 2025 (SEI 4.4/5.0)
WVU	**Reinforcement Learning and Control** (graduate), Spring 2021 (SEI 4.6/5.0)

Courses Taught

RIT	**System Dynamics**, Fall 2023, Spring 2024
WVU	**Automatic Control**, Summer 2021, 2022 (SEI 4.7/5.0)

Research Advising and Mentoring

2024–present	Chirayu Salgarkar, Ph.D. student
2021–2024	Lunet Yifru, M.S. student
2021–2024	Joshua Yancosek, M.S. student
2023–2024	Aniket Narendra Patil, M.S. student (co-advised with Prof. Cecilia Alm)
2023–2024	Kaustubh Gaikwad, M.S. student (co-advised with Prof. Cecilia Alm)
2021, 2022	Mentor, NSF REU Site at West Virginia University

Ph.D. Thesis Committees

2023	Rogerio Rodrigues Lima, West Virginia University
2022	Robert Tempke, West Virginia University
2021	Jared Strader, West Virginia University

Professional Service

NSF Panels	National Science Foundation review panels: DCSD (2021, 2023, 2024), CPS (2022), SLES (2023, ×2), FMitF (2024), SBIR/STTR (2024)
	Program Committee, AAAI Conference on Artificial Intelligence (AAAI), 2027
	Program Committee, “Foundation Models for Decision Making” workshop, NeurIPS, 2023, 2024
	Session Chair, American Control Conference (ACC), 2026
	Session Chair, European Control Conference (ECC), 2026
2023–present	Senior Personnel, NSF AWARE-AI NRT at RIT (co-lead, Software track)
	Co-organizer, “Machine Learning for Autonomous Driving (ML4AD)” workshop, NeurIPS, 2021, 2022
	Co-organizer, “Fault Diagnosis for Safety-Critical Autonomous Spacecraft Systems” workshop, Mid-Atlantic Space Grant Data Science Consortium (NASA)
	Co-organizer, “Robotics Seminar Series,” West Virginia University

Reviewer Service

Journals	IEEE Transactions on Intelligent Vehicles; IEEE Transactions on Vehicular Technology; IEEE Robotics and Automation Letters; IEEE Transactions on Aerospace and Electronic Systems; Journal of Aerospace Information Systems; Sustainable Energy Technologies and Assessments; Energies
Conferences	NeurIPS; ICLR; IROS; Robotics: Science and Systems (RSS); IEEE Conference on Decision and Control (CDC); American Control Conference (ACC); European Control Conference (ECC); IEEE Intelligent Vehicles Symposium (IV); IEEE Intelligent Transportation Systems Conference (ITSC); ASME DSCC

Professional Memberships

2024–present	Member, ASME
2023–present	Member, IEEE